



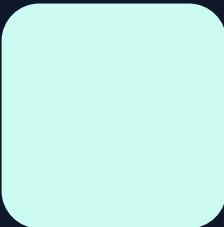
DAY 5



Full Evaluation & Explainability

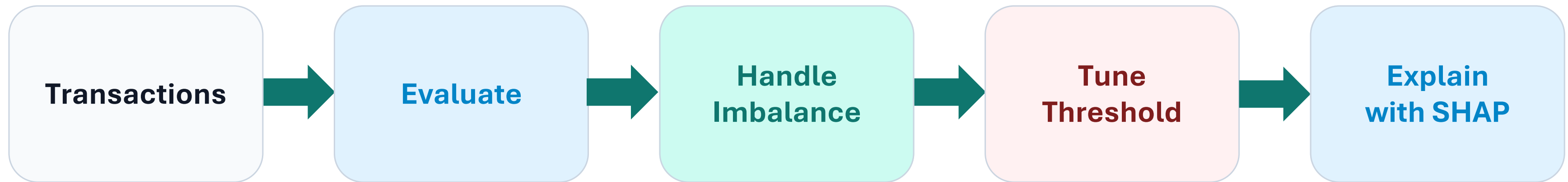
Credit Card Fraud Detection — Imbalance, Threshold Tuning & SHAP

Week 8 • AI & ML Internship



Prepared By Sadeel Ardah

Prediction is not the end of the workflow



Goal

Use the right metrics, improve minority-class learning, choose a useful threshold, and explain the model.

Fraud is rare — and that changes how we evaluate

Majority Class

Normal transactions
Very large number of examples

Minority Class

Fraud transactions
Very small number of examples

A model can look accurate while missing the cases we care about.

Accuracy alone is not enough

Precision

When predicted Fraud,
how often correct?

Recall

Of all real Fraud,
how many caught?

F1

Balance between
Precision & Recall

ROC-AUC

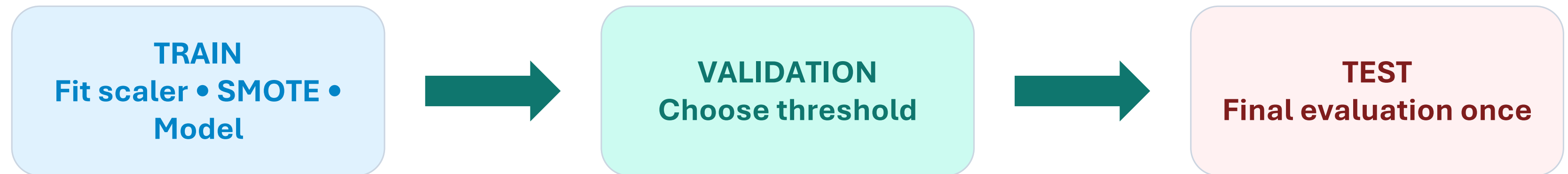
Ranking quality
across thresholds

PR-AUC

Positive-class quality
under imbalance

The metric choice should match the cost of the mistakes.

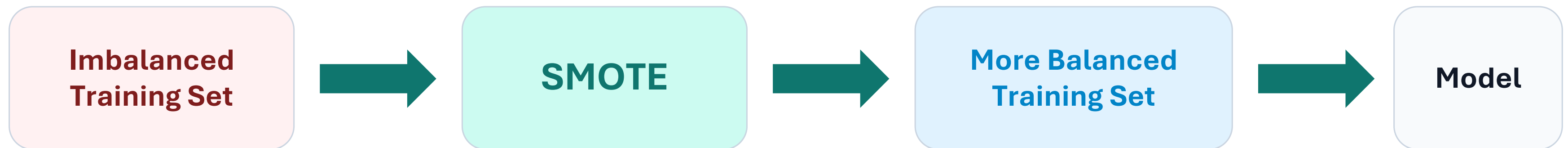
Keep tuning separate from final evaluation



Why?

Choosing the threshold on the test set would make the final result optimistically biased.

SMOTE creates synthetic minority examples in training data



Critical Rule

SMOTE is applied to TRAINING DATA ONLY. Validation and test data stay real and untouched.

Catching more fraud can create more false alarms

Lower Threshold

More transactions become Fraud predictions

- ↑ Recall
- ↓ Precision can happen

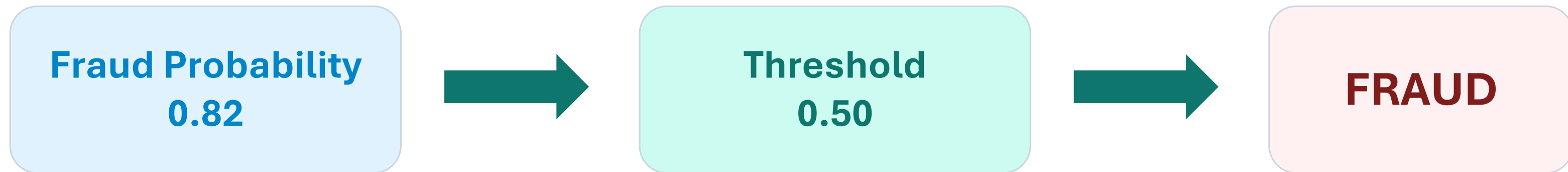
Higher Threshold

Fewer transactions become Fraud predictions

- ↑ Precision can happen
- ↓ Recall

There is no universally perfect threshold.

Probability becomes a class through a decision threshold



Professional workflow

Tune the threshold on the validation set, then evaluate that fixed threshold on the untouched test set.

Three stages answer three different questions

1. Baseline

How does the model behave on the original imbalanced training data?

2. SMOTE

Does minority-class learning improve after balancing the training data?

3. Tuned Threshold

Can we choose a better operating point without retraining the model?

Compare Precision • Recall • F1 • ROC-AUC • PR-AUC

SHAP explains how features contribute to model output



Important

SHAP explains the model's behavior; it does not prove real-world causation.

Two levels of explainability

GLOBAL EXPLANATION

Question:
Which features matter most across many predictions?

Output:
Feature importance / SHAP summary

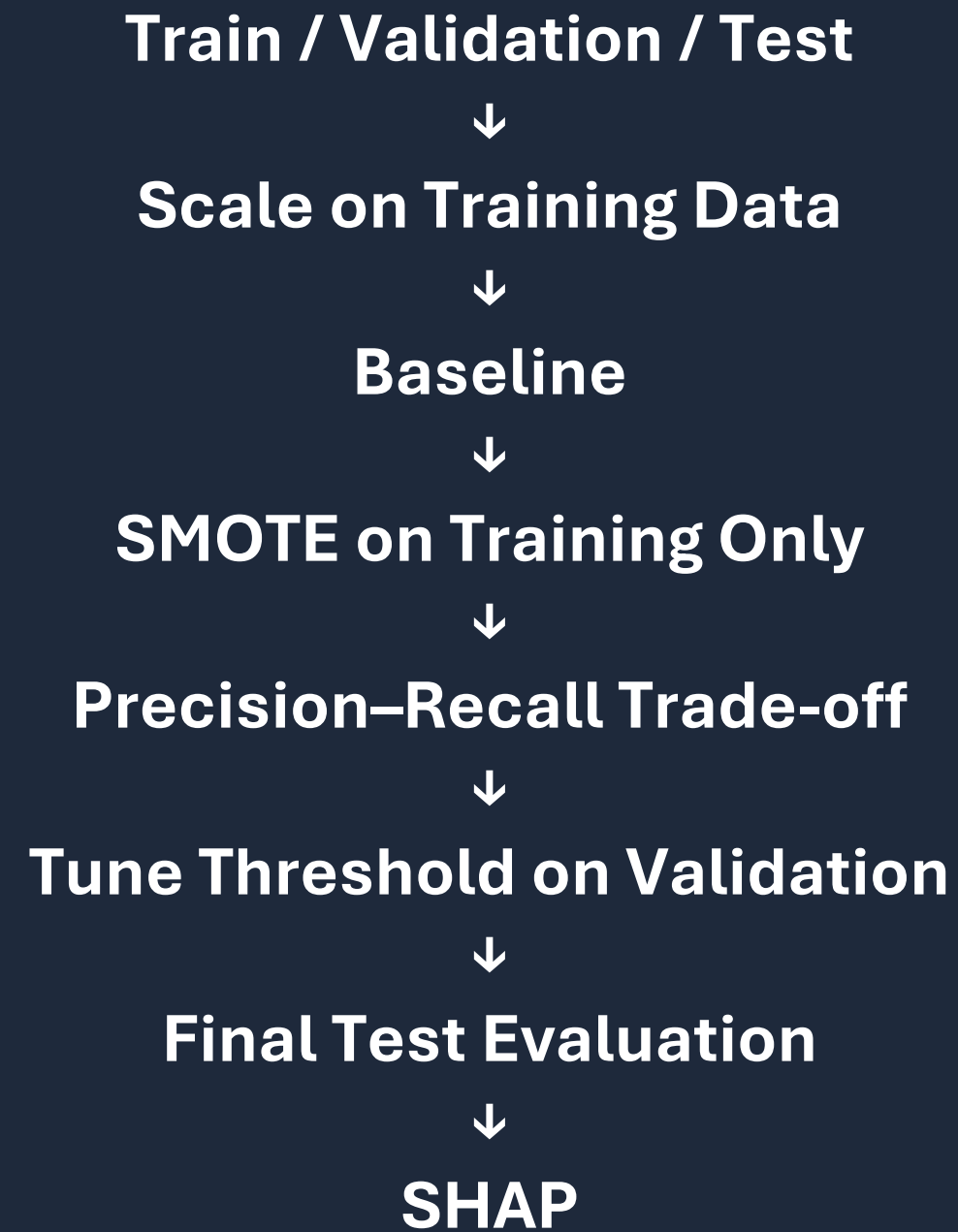
LOCAL EXPLANATION

Question:
Why did this specific transaction receive this prediction?

Output:
SHAP waterfall / feature contributions

Global = overall behavior • Local = one prediction

Day 5 Mental Map



KEY TAKEAWAY

Use the right metrics.
Handle imbalance without leakage.
Tune the threshold correctly.
Explain the prediction.

What I Learned Today

- 1 Why accuracy can mislead on imbalanced data
- 2 Precision, Recall, F1, ROC-AUC and PR-AUC
- 3 How SMOTE balances training data
- 4 Why SMOTE must not touch validation/test data
- 5 The Precision–Recall trade-off
- 6 How decision thresholds change predictions
- 7 Why threshold tuning belongs on validation data
- 8 How SHAP explains feature contributions
- 9 Global vs. local model explanations

FINAL IDEA

**A strong model is
not just accurate.**

**It must be
evaluated correctly
and its decisions
should be
explainable**